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Theory of General Perturbations for Non-Canonical Systems

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Abstract

The theory of general perturbations originally developed for the solution of canonical systems is generalized so as to be applicable to non-canonical systems. The features of the original theory are preserved as far as possible. Perturbed Kepler motion is discussed as an application of the general theory. Another application to the non-linear oscillations is shown by the solution of van del Pol's equation. Finally the present theory is compared with the original theory applied to the generalized canonical system.

Key words: General perturbations; Non-canonical system; Non-linear oscillations.

1. *Introduction*

The given equations of motion

$$\frac{dz_j}{dt}=Z_j(z)\,, \qquad (j=1,\cdots,m)\,, \tag{1}$$

are transformed, by the change of variables $z\to\zeta$, to

$$\frac{d\zeta_j}{dt}=Z_j^*(\zeta)\,, \qquad (j=1,\cdots,m)\,, \tag{2}$$

in such a way that the new equations of motion (2) have some advantages for the solution. This is the idea of perturbation theories based on the transformations of variables.

When the given equations of motion are in the canonical form

$$\frac{dx_j}{dt}=\frac{\partial F}{\partial y_j},\ \frac{dy_j}{dt}=-\frac{\partial F}{\partial x_j}\,, \qquad (j=1,\cdots,m')\,, \tag{3}$$

with the Hamiltonian $F(x, y)$, we apply a canonical transformation $x, y\to\xi, \eta$ so that the new equations are again canonical and may be written in the form

$$\frac{d\xi_j}{dt}=\frac{\partial F^*}{\partial \eta_j},\ \frac{d\eta_j}{dt}=-\frac{\partial F^*}{\partial \xi_j}\,, \qquad (j=1,\cdots,m')\,, \tag{4}$$

where we have

$$F(x, y)=F^*(\xi, \eta)\,. \tag{5}$$

Several theories (VON ZEIPEL 1916; HORI 1966; DEPRIT 1969) have been developed for the solution of the canonical systems, and in all of them, equation (5)

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is the basic relation. In general, if either the given system or the applied transformation is not canonical, we do not have such a simple relation as equation (5) linking the old and the new systems, and this yields some complications for the construction of perturbation theories for non-canonical systems.

However, a trial was made by BROWER and HORI (1961) with some success, by applying a canonical transformation to a non-canonical system—the drag problem of the motion of an artificial satellite. Later the theory was improved by LANE (1965).

We present here the generalization of the HORI's (1966) version of the perturbation theory originally developed for canonical systems. The generalization based on the DEPRIT's (1969) version has been made by KAMEL (1970). On the other hand, BREEDLOVE (1969), CHOI (1970), and KAMEL (1971) have adopted an entirely different approach and reformulated equations (1) in the canonical form

$$\frac{dz_j}{dt}=\frac{\partial F}{\partial p_j},\quad \frac{dp_j}{dt}=-\frac{\partial F}{\partial z_j}\,,\qquad (j=1,\cdots,m)\,, \tag{6}$$

with the Hamiltonian

$$F=\sum_{j=1}^{m} p_j Z_j\,, \tag{7}$$

by artificially introducing m momenta p_j conjugate to z_j. The original order m of equations (1) is doubled in equations (6), and this is the price paid for obtaining canonicity. In Section 8 we compare this approach with the present one, and show that the former reduces to the latter if the algorithm is required to be independent of artificial momenta p.

2. *Generalization*

In the HORI's (1966) version of the perturbation theory, use is made of the canonical transformation $x, y\to\xi, \eta$,

$$x_j=\xi_j+\sum_{n=1}\frac{1}{n!}D_S^n\xi_j\,,\quad y_j=\eta_j+\sum_{n=1}\frac{1}{n!}D_S^n\eta_j\,, \tag{8}$$

where the operator D_S is defined by

$$D_S f(\xi,\eta)=\{f(\xi,\eta), S(\xi,\eta)\}\,, \tag{9}$$

the braces standing for the Poisson bracket expression. For an arbitrary function $f(x, y)$, we have the expansion formula

$$f(x,y)=f(\xi,\eta)+\sum_{n=1}\frac{1}{n!}D_S^n f(\xi,\eta)\,. \tag{10}$$

We see that equation (10) includes equations (8) as special cases. In these equations, we assume that the function $S(\xi, \eta)$, which generates the canonical transformation, is small enough so as to guarantee the convergence of the infinite series in the right-hand members of equations (8) and (10).

We first generalize the canonical transformation (8) and consider a general transformation $z\to\zeta$ by introducing a set of m generating functions $T_j(\zeta)$ of

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variables $\zeta_1, \cdots, \zeta_m$ in place of one function $S(\xi, \eta)$ in the canonical case. Let the operator D_T be defined by

$$D_T f(\zeta) = T_j \frac{\partial f}{\partial \zeta_j}, \quad \text{and} \quad D_T^n f = D_T^{n-1}\left(T_j \frac{\partial f}{\partial \zeta_j}\right), \tag{11}$$

where the Einstein convention is used. This operator is reduced to D_S if m is even, say $2m'$, and we put $\zeta_j = \xi_j$, $\zeta_{m'+j} = \eta_j$ $(j=1, \cdots, m')$, T_j and S being related by

$$T_j = \frac{\partial S}{\partial \eta_j}, \quad T_{m'+j} = -\frac{\partial S}{\partial \xi_j}, \qquad (j=1, \cdots, m') . \tag{12}$$

Equations (8) are then generalized to

$$z_j = \zeta_j + \sum_{n=1} \frac{1}{n!} D_T^n \zeta_j = e^{D_T} \zeta_j , \tag{13}$$

and equation (10) to

$$f(z) = f(\zeta) + \sum_{n=1} \frac{1}{n!} D_T^n f(\zeta) = e^{D_T} f(\zeta) . \tag{14}$$

We assume that T_j are small enough to assure the convergence of the series.

The inverse transformation of equation (10), which expresses $f(\xi, \eta)$ in terms of x, y, is given by

$$f(\xi, \eta) = f(x, y) + \sum_{n=1} \frac{(-1)^n}{n!} D_{\bar{S}}^n f(x, y) , \tag{15}$$

where $\bar{S}$ stands for $[S(\xi, \eta)]_{\xi, \eta \to x, y}$. Similarly, if $\bar{T}_j$ and $D_{\bar{T}}$ are introduced by

$$\bar{T}_j = [T_j(\zeta)]_{\zeta \to z}, \quad D_{\bar{T}} = \bar{T}_j \frac{\partial}{\partial z_j} , \tag{16}$$

we have the inverse relation of equation (14),

$$f(\zeta) = f(z) + \sum_{n=1} \frac{(-1)^n}{n!} D_{\bar{T}}^n f(z) = e^{-D_{\bar{T}}} f(z) , \tag{17}$$

corresponding to equation (15) in the canonical case.

3. *New Equations of Motion*

The new equations of motion (2) are derived from equations (1) by three steps: we first express ζ in terms of z, as the special case of equation (17) with $f(z) = z_j$,

$$\zeta_j = e^{-D_T} z_j ; \tag{18}$$

we then take the time derivative of equation (18), and in view of

$$\frac{d}{dt} = Z_j(z) \frac{\partial}{\partial z_j} \equiv D_{\bar{Z}} , \tag{19}$$

we have

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$$\frac{d\zeta_j}{dt}=D_{\bar{Z}}e^{-D_{\bar{T}}}z_j : \tag{20}$$

and finally we apply the formula (14) to the right-hand member of equation (20) so as to express $d\zeta_j/dt$ in terms of ζ. The result is

$$\frac{d\zeta_j}{dt}=e^{D_T}D_Ze^{-D_T}\zeta_j=Z_j{}^*(\zeta) . \tag{21}$$

We note that the T_j are small quantities, say, of the first order, and the terms of the middle member are arranged according to their order of magnitude:

$$e^{D_T}D_Ze^{-D_T}=\sum\frac{1}{\mu!}D_T^{\mu}D_Z\sum\frac{(-1)^{\nu}}{\nu!}D_T^{\nu}=\sum_{n=0}\sum_{\nu=0}^{n}\frac{(-1)^{\nu}}{(n-\nu)!\nu!}D_T^{n-\nu}D_ZD_T^{\nu} . \tag{22}$$

The order n terms are written in the form

$$\frac{1}{n!}\sum_{\nu=0}^{n}(-1)^{\nu}\binom{n}{\nu}D_T^{n-\nu}D_ZD_T^{\nu} . \tag{23}$$

If the generalized Poisson bracket expression is introduced by

$$[Z]_j=Z_j , \tag{24}$$

$$\mathscr{D}_TZ_j=[Z, T]_j=T_k\frac{\partial Z_j}{\partial\zeta_k}-Z_k\frac{\partial T_j}{\partial\zeta_k} , \tag{25}$$

and

$$\mathscr{D}_T^nZ_j=[\cdots[[Z, T], T], \cdots T]_j=T_k\frac{\partial}{\partial\zeta_k}\mathscr{D}_T^{n-1}Z_j-(\mathscr{D}_T^{n-1}Z_k)\frac{\partial T_j}{\partial\zeta_k} , \tag{26}$$

then we can prove that

$$\sum_{\nu=0}^{n}(-1)^{\nu}\binom{n}{\nu}D_T^{n-\nu}D_ZD_T^{\nu}=(\mathscr{D}_T^nZ_k)\frac{\partial}{\partial\zeta_k}=D_{[\cdots[[Z, T], T], \cdots T]} . \tag{27}$$

In fact, for $n=0$, equation (27) reduces to

$$D_Z=D_{[Z]} ,$$

which is true because of equation (24). Suppose equation (27) is correct for $n=N-1$. Then we have

$$\begin{aligned}
&\sum_{\nu=0}^{N}(-1)^{\nu}\binom{N}{\nu}D_T^{N-\nu}D_ZD_T^{\nu}\\
&=D_T\sum_{\nu=0}^{N-1}(-1)^{\nu}\binom{N-1}{\nu}D_T^{N-1-\nu}D_ZD_T^{\nu}-\sum_{\nu=0}^{N-1}(-1)^{\nu}\binom{N-1}{\nu}D_T^{N-1-\nu}D_ZD_T^{\nu}D_T\\
&=D_TD_{[\cdots[Z, T], \cdots T]}-D_{[\cdots[Z, T], \cdots T]}D_T\\
&=T_k\frac{\partial}{\partial\zeta_k}\left[(\mathscr{D}_T^{N-1}Z_l)\frac{\partial}{\partial\zeta_l}\right]-(\mathscr{D}_T^{N-1}Z_k)\frac{\partial}{\partial\zeta_k}\left(T_l\frac{\partial}{\partial\zeta_l}\right)\\
&=\left[T_k\frac{\partial}{\partial\zeta_k}\mathscr{D}_T^{N-1}Z_l-(\mathscr{D}_T^{N-1}Z_k)\frac{\partial T_l}{\partial\zeta_k}\right]\frac{\partial}{\partial\zeta_l}
\end{aligned}$$

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$$=(\mathscr{D}_T^N Z_l)\frac{\partial}{\partial\zeta_l}=D_{[\cdots[Z,T],T],\cdots T]}\,.$$

Equation (27) is thus proved. Therefore we have

$$e^{D_T}D_Z e^{-D_T}\zeta_j=\left(D_{[Z]}+D_{[Z,T]}+\frac{1}{2!}D_{[[Z,T],T]}+\cdots\right)\zeta_j$$
$$=Z_j+[Z,T]_j+\frac{1}{2!}[[Z,T],T]_j+\cdots=e^{\mathscr{D}_T}Z_j\,. \tag{28}$$

When equations (1) are equivalent to equations (3) with $m=2m'$, we have

$$\left.\begin{aligned}&[\cdots[[Z,T],T],\cdots T]_j=\frac{\partial}{\partial\eta_j}\{\cdots\{\{F,S\},S\},\cdots S\}\,,\\&[\cdots[[Z,T],T],\cdots T]_{m'+j}=-\frac{\partial}{\partial\xi_j}\{\cdots\{\{F,S\},S\},\cdots S\}\,,\end{aligned}\right\}(j=1,\cdots,m')\,, \tag{29}$$

as is easily verified, where

$$Z_j(\zeta)=\frac{\partial}{\partial\eta_j}F(\xi,\eta)\,,\qquad Z_{m'+j}(\zeta)=-\frac{\partial}{\partial\xi_j}F(\xi,\eta)\,, \tag{30}$$

in addition to equation (12).

The operators D_T for a scalar $f(\zeta)$ and $\mathscr{D}_T$ for a vector $Z_j(\zeta)$ are called Lie derivatives after LIE (1888), and we call equation (14) Lie's expansion formula. The author does not know whether equation (27) and therefore equation (28) are new or not. In Section 8, we see another indirect proof of equation (28), which was suggested by H. Kinoshita of the Tokyo Astronomical Observatory.

4. *Algorithm through the Fourth Order*

We assume that Z_j, T_j, and Z_j^* may be expanded in powers of a small parameter of the first order,

$$Z_j=Z_j^{(0)}+Z_j^{(1)}+Z_j^{(2)}+Z_j^{(3)}+\cdots\,, \tag{31}$$

$$T_j=T_j^{(1)}+T_j^{(2)}+T_j^{(3)}+\cdots\,, \tag{32}$$

$$Z_j^*=Z_j^{*(0)}+Z_j^{*(1)}+Z_j^{*(2)}+Z_j^{*(3)}+\cdots\,, \tag{33}$$

where $T_j^{(0)}$ is absent in equation (32) because T_j is assumed to be of the first order.

We have from equation (21), in view of equations (28), (31), (32), (33), the following terms:

Zeroth order: $$Z_j^{(0)}=Z_j^{*(0)}\,. \tag{34}$$

First order: $$[Z^{(0)},T^{(1)}]_j+Z_j^{(1)}=Z_j^{*(1)}\,. \tag{35}$$

Second order: $$[Z^{(0)},T^{(2)}]_j+[Z^{(1)},T^{(1)}]_j+\frac{1}{2}[[Z^{(0)},T^{(1)}],T^{(1)}]_j+Z_j^{(2)}=Z_j^{*(2)}\,. \tag{36}$$

Third order: $$[Z^{(0)},T^{(3)}]_j+[Z^{(1)},T^{(2)}]_j+[Z^{(2)},T^{(1)}]_j+\frac{1}{2}[[Z^{(0)},T^{(1)}],T^{(2)}]_j$$

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$$+\frac{1}{2}[[Z^{(0)}, T^{(2)}], T^{(1)}]_j+\frac{1}{2}[[Z^{(1)}, T^{(1)}], T^{(1)}]_j$$
$$+\frac{1}{6}[[[Z^{(0)}, T^{(1)}], T^{(1)}], T^{(1)}]_j+Z_j^{(3)}=Z_j^{*(3)} . \tag{37}$$

Fourth order: $[Z^{(0)}, T^{(4)}]_j+[Z^{(1)}, T^{(3)}]_j+[Z^{(2)}, T^{(2)}]_j+[Z^{(3)}, T^{(1)}]_j$

$$+\frac{1}{2}[[Z^{(0)}, T^{(1)}], T^{(3)}]_j+\frac{1}{2}[[Z^{(0)}, T^{(2)}], T^{(2)}]_j+\frac{1}{2}[[Z^{(0)}, T^{(3)}], T^{(1)}]_j$$
$$+\frac{1}{2}[[Z^{(1)}, T^{(1)}], T^{(2)}]_j+\frac{1}{2}[[Z^{(1)}, T^{(2)}], T^{(1)}]_j+\frac{1}{2}[[Z^{(2)}, T^{(1)}], T^{(1)}]_j$$
$$+\frac{1}{6}[[[Z^{(0)}, T^{(1)}], T^{(1)}], T^{(2)}]_j+\frac{1}{6}[[[Z^{(0)}, T^{(1)}], T^{(2)}], T^{(1)}]_j$$
$$+\frac{1}{6}[[[Z^{(0)}, T^{(2)}], T^{(1)}], T^{(1)}]_j+\frac{1}{6}[[[Z^{(1)}, T^{(1)}], T^{(1)}], T^{(1)}]_j$$
$$+\frac{1}{24}[[[[Z^{(0)}, T^{(1)}], T^{(1)}], T^{(1)}], T^{(1)}]_j+Z_j^{(4)}=Z_j^{*(4)} . \tag{38}$$

These equations correspond to equations (16) to (20) of the previous paper on the canonical case.

As in the canonical case, equations for each order, are simplified with the use of the equations of the preceding orders, and we have:

Zeroth order: $$Z_j^{(0)}=Z_j^{*(0)} . \tag{34'}$$

First order: $$[Z^{(0)}, T^{(1)}]_j+Z_j^{(1)}=Z_j^{*(1)} . \tag{35'}$$

Second order: $$[Z^{(0)}, T^{(2)}]_j+\frac{1}{2}[Z^{(1)}+Z^{*(1)}, T^{(1)}]_j+Z_j^{(2)}=Z_j^{*(2)} . \tag{36'}$$

Third order: $$[Z^{(0)}, T^{(3)}]_j+\frac{1}{2}[Z^{(1)}+Z^{*(1)}, T^{(2)}]_j+\frac{1}{2}[Z^{(2)}+Z^{*(2)}, T^{(1)}]_j$$
$$+\frac{1}{12}[[Z^{(1)}-Z^{*(1)}, T^{(1)}], T^{(1)}]_j+Z_j^{(3)}=Z_j^{*(3)} . \tag{37'}$$

Fourth order: $$[Z^{(0)}, T^{(4)}]_j+\frac{1}{2}[Z^{(1)}+Z^{*(1)}, T^{(3)}]_j+\frac{1}{2}[Z^{(2)}+Z^{*(2)}, T^{(2)}]_j$$
$$+\frac{1}{2}[Z^{(3)}+Z^{*(3)}, T^{(1)}]_j+\frac{1}{12}[[Z^{(1)}-Z^{*(1)}, T^{(1)}], T^{(2)}]_j$$
$$+\frac{1}{12}[[Z^{(1)}-Z^{*(1)}, T^{(2)}], T^{(1)}]_j+\frac{1}{12}[[Z^{(2)}-Z^{*(2)}, T^{(1)}], T^{(1)}]_j$$
$$+Z_j^{(4)}=Z_j^{*(4)} . \tag{38'}$$

These correspond to equations (16′) to (20′) of the previous paper and have analogous forms.

The m equations in each order contain $2m$ unknowns, $T_j^{(n)}$ and $Z_j^{*(n)}$ ($j=1, \cdots, m$), so that some principle should be introduced to determine $2m$ unknowns from m equations.

Let us consider the auxiliary equations

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$$\frac{d\zeta_j}{d\tau}=Z_j^{*(0)}(\zeta)\,, \qquad (j=1,\cdots,m)\,, \tag{39}$$

which are equivalent, except for the notation, to the unperturbed version,

$$\frac{dz_j}{dt}=Z_j^{(0)}(z)\,,$$

of the given equations of motion (1). We assume that the unperturbed problem is solvable, and denote its solution by

$$\zeta_j=\zeta_j(\tau, c_1, \cdots, c_m)\,, \qquad (j=1,\cdots,m)\,. \tag{40}$$

The term $[Z^{(0)}, T^{(n)}]_j$ which involves the unknouns $T_j^{(n)}$ in the n-th order equation is written, in view of equations (39), in the form

$$[Z^{(0)}, T^{(n)}]_j=T_k^{(n)}\frac{\partial Z_j^{(0)}}{\partial\zeta_k}-Z_k^{(0)}\frac{\partial T_j^{(n)}}{\partial\zeta_k}=-\frac{dT_j^{(n)}}{d\tau}+\frac{\partial Z_j^{(0)}}{\partial\zeta_k}T_k^{(n)}\,, \tag{41}$$

and we note that the coefficients $\partial Z_j^{(0)}/\partial\zeta_k$ of $T_k^{(n)}$ are known functions of τ because of equations (40). Therefore the equations for $T_j^{(n)}$ in each order are

$$-\frac{dT_j^{(n)}}{d\tau}+Z_{j\zeta_k}^{(0)}(\tau)\cdot T_k^{(n)}+\Phi_j^{(n)}(\tau)=Z_j^{*(n)}\,, \qquad (j=1,\cdots,m)\,, \tag{42}$$

where $\Phi_j^{(n)}$ are known functions of τ, determined in the preceding orders of approximation.

Equations (42) reduce to the single equation

$$-\frac{dS_n}{d\tau}+\Psi_n(\tau)=F_n^* \tag{43}$$

in the canonical case. S_n is then obtained by the quadrature

$$S_n=\int(\Psi_n-F_n^*)d\tau\,. \tag{44}$$

In the present case, the $T_j^{(n)}$ are obtained by solving the non-homogeneous linear differential equations with variable coefficients

$$\frac{dT_j^{(n)}}{d\tau}-Z_{j\zeta_k}^{(0)}(\tau)\cdot T_k^{(n)}=\Phi_j^{(n)}-Z_j^{*(n)}\,, \qquad (j=1,\cdots,m)\,. \tag{45}$$

As F_n^* is so determined in equation (44) that S_n is free from secular or mixed secular terms, $Z_j^{*(n)}$ in equations (45) are likewise determined so that $T_j^{(n)}$ are free from these terms. This is necessary in the determination of $Z_j^{*(n)}$ and $T_j^{(n)}$ in so far as we require that our perturbation theory be formally valid for an unlimited interval of time.

In the canonical case, the choice of the new Hamiltonian as the average of Ψ_n in τ, $F_n^*=\langle\Psi_n\rangle$, not only meets the demand for S_n but also yields the integral $F_0(\xi,\eta)=$const. In the present case, the demand for $T_j^{(n)}$ is generally not sufficient to determine $Z_j^{*(n)}$ such that the new equations of motion become more tractable by admitting some integrals, or in other ways. Analogous to the condition

$$\{F_0, F^*\}=0 \tag{46}$$

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in the canonical case, we here put

$$[Z^{(0)}, Z^*]_j=0\,, \qquad (j=1, \cdots, m)\,, \tag{47}$$

as the tractability condition, which leads to

$$\frac{dZ_j^{(0)}}{dt}=Z_k^*\frac{\partial Z_j^{(0)}}{\partial \zeta_k}=Z_k^{(0)}\frac{\partial Z_j^*}{\partial \zeta_k}=\frac{dZ_j^*}{d\tau}\,, \tag{48}$$

or

$$\frac{dZ_j^{*(n)}}{d\tau}=\frac{\partial Z_j^{(0)}}{\partial \zeta_k}Z_k^{*(n)}\,, \qquad (j=1, \cdots, m)\,, \tag{49}$$

in each order. In fact, equation (48) shows that if the Z_j^* depend only on the integration constants of the solution of the auxiliary equations and not on τ for some indices j, we have, for such j, the integrals

$$Z_j^{(0)}=\text{const.} \tag{50}$$

This is the case of perturbed elliptic motion discussed in Section 6. In another case, condition (49) leads to the form $Z_j^*=f(\zeta_1^2+\zeta_2^2)\zeta_j$ $(j=1, 2)$, and the new equations become integrable, as shown in Section 7. Generally, the tractability depends on the form of $Z_j^{(0)}$. We note that condition (49) is also derived when the original canonical theory is applied to the generalized canonical system with the requirement that the solution $z(t)$ is obtained without any information about the momenta p which are artificially introduced (see Section 8). Finally, we note that conditions (47) reduce to

$$\frac{\partial}{\partial \eta_j}\{F_0, F^*\}=0, \quad \frac{\partial}{\partial \xi_j}\{F_0, F^*\}=0\,, \qquad (j=1, \cdots, m')\,,$$

if equations (2) are written in the form (4) with $m=2m'$. Thus equations (47) are equivalent to equations (46) of the canonical case.

5. *Perturbed Elliptic Motion*

In equations (1) with $Z_j=Z_j^{(0)}+Z_j^{(1)}$, we replace the variables $z_1, z_2, \cdots, z_6$ by the Delaunay variables L, G, H, l, g, h. Further, we put $Z_j^{(0)}=0$ for $j=1, 2, 3, 5, 6$ and $Z_4^{(0)}=\mu^2/z_1^3$. Equations (1) then give a perturbed elliptic motion with $Z_j^{(1)}$ as the disturbing force components.

In this case, equations (41) become

$$[Z^{(0)}, T^{(n)}]_j=-\frac{dT_4^{(n)}}{d\tau}\,, \qquad (j=1, 2, 3, 5, 6)\,, \tag{51}$$

$$[Z^{(0)}, T^{(n)}]_4=-\frac{dT_4^{(n)}}{d\tau}-\frac{3\mu^2}{\zeta_1^4}T_1^{(n)}\,, \tag{52}$$

and equations (35′) for the first order lead to

$$-\frac{dT_j^{(1)}}{d\tau}+Z_j^{(1)}=Z_j^{*(1)}\,, \qquad (j=1, 2, 3, 5, 6)\,, \tag{53}$$

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$$-\frac{dT_4^{(1)}}{d\tau}-\frac{3\mu^2}{\zeta_1^4}T_1^{(1)}+Z_4^{(1)}=Z_4^{*(1)}\,, \tag{54}$$

where, as the auxiliary equations (39) show, ζ_1, ζ_2, ζ_3, ζ_5 and ζ_6 are constant in τ while ζ_4 is a linear function of τ, $\zeta_4=(\mu^2/\zeta_1^3)\tau+\text{const.}$

In equations (53) and (54), $Z_j^{(1)}$ $(j=1,\cdots,6)$ depend on τ through ζ_4. It is assumed that $Z_j^{(1)}$ are expressed as Fourier series in τ. We then have

$$Z_j^{*(1)}=\langle Z_j^{(1)}\rangle\,, \qquad (j=1,2,3,5,6)\,, \tag{55}$$

$$T_j^{(1)}=\int(Z_j^{(1)}-Z_j^{*(1)})d\tau=\frac{\zeta_1^3}{\mu^2}\int(Z_j^{(1)}-Z_j^{*(1)})d\zeta_4\,, \qquad (j=1,2,3,5,6)\,, \tag{56}$$

and

$$Z_4^{*(1)}=\left\langle Z_4^{(1)}-\frac{3\mu^2}{\zeta_1^4}T_1^{(1)}\right\rangle, \tag{57}$$

$$T_4^{(1)}=\frac{\zeta_1^3}{\mu^2}\int\left(Z_4^{(1)}-\frac{3\mu^2}{\zeta_1^4}T_1^{(1)}-Z_4^{*(1)}\right)d\zeta_4\,. \tag{58}$$

Therefore, we have the unique determination of $Z_j^{*(1)}$ and $T_j^{(1)}$ except for the constants of integration in $T_j^{(1)}$, and the situation does not change in the second and higher orders.

As equations (55) and (57) show, τ or ζ_4 is eliminated from $Z_j^{*(1)}$, and in general from $Z_j^{*(2)}$, $Z_j^{*(3)}$, $\cdots$ if we go to higher orders. In the new system (2), therefore, the equation for ζ_4,

$$\frac{d\zeta_4}{dt}=\frac{\mu^2}{\zeta_1^3}+Z_4^{*(1)}+Z_4^{*(2)}+\cdots\,, \tag{59}$$

is separated from the others

$$\frac{d\zeta_j}{dt}=Z_j^{*(1)}+Z_j^{*(2)}+\cdots\,, \qquad (j=1,2,3,5,6)\,. \tag{60}$$

If we can solve ζ_1, ζ_2, ζ_3, ζ_5, and ζ_6 of equations (60), ζ_4 is obtained from equation (59) by a quadrature. Therefore the order of the equations of motion is reduced by one by means of the transformation $z\rightarrow\zeta$ given by equations (13), where T_j are given by equations (56), (58), and similar equations in higher orders. The new equations of motion (60), together with equation (59), determine the long range behavior of the variables ζ. A second transformation $\zeta_j\rightarrow\zeta_j'$ $(j=1,2,3,5,6)$ is used for the solution of equations (60). In this case, the $Z_j^{*(1)}$ play the roll of $Z_j^{(0)}$, that is, the undisturbed parts of the force components, and $Z_j^{*(2)}+\cdots$ become the disturbing force components. Except for this, the procedure of Section 4 is again applied to equations (60). Thus we may determine $T_j^{*(1)}$, $T_j^{*(2)}$, $\cdots$ and $Z_j^{**(1)}$, $Z_j^{**(2)}$, $\cdots$ for $j=1,2,3,5,6$.

We note that the right-hand member of equation (59) is expressed in terms of ζ_1', ζ_2', ζ_3', ζ_5', and ζ_6' by the help of $T_j^*(\zeta_1',\zeta_2',\zeta_3',\zeta_5',\zeta_6')$ $(j=1,2,3,5,6)$, T_4^* not being needed since ζ_4' is absent.

6. *Main Problem of the Motion of an Artificial Satellite*

We consider, as an example, the main problem of the motion of an artificial

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satellite. The problem has been discussed with the use of von Zeipel's method (BROUWER 1959; KOZAI 1962), or with the use of the canonical version of the present method (HORI 1966; AKSNES 1970). Observing that the disturbing forces $Z_j^{(1)}$, $Z_j^{(2)}$ are independent of z_6, and that z_3 is constant and considered as a parameter, we solve the fourth order system

$$\frac{dz_j}{dt}=Z_j^{(0)}+Z_j^{(1)}+Z_j^{(2)}\,, \qquad (j=1,2,4,5)\,, \tag{61}$$

where

$$Z_j^{(0)}=0\ (j=1,2,5)\,, \quad Z_4^{(0)}=\frac{\mu^2}{z_1^3}\,, \tag{62}$$

and

$$Z_j^{(i)}=\frac{\partial F_i}{\partial z_{3+j}}\,, \quad Z_{3+j}^{(i)}=-\frac{\partial F_i}{\partial z_j}\,, \qquad (i,j=1,2)\,, \tag{63}$$

with the disturbing function F_1+F_2.

Equations (55) to (58) then give

$$Z_1{}^{*(1)}=\langle F_{1\zeta_4}(\zeta_1,\zeta_2,z_3,\zeta_4,\zeta_5,-)\rangle=0\,, \tag{64}$$

$$T_1{}^{(1)}=\frac{1}{n}\int\frac{\partial F_1}{\partial\zeta_4}d\zeta_4=\frac{1}{n}F_{1p}\,, \qquad \left(n=\frac{\mu^2}{\zeta_1^3}\,,\ F_{1p}=F_1-\langle F_1\rangle\right), \tag{65}$$

$$Z_2{}^{*(1)}=\left\langle\frac{\partial F_1}{\partial\zeta_5}\right\rangle=0\,, \tag{66}$$

$$T_2{}^{(1)}=\frac{1}{n}\int\frac{\partial F_1}{\partial\zeta_5}d\zeta_4\,, \tag{67}$$

$$Z_4{}^{*(1)}=-\left\langle\frac{\partial F_1}{\partial\zeta_1}\right\rangle=-\frac{\partial F_1{}^*}{\partial\zeta_1}\,, \qquad (F_1{}^*=\langle F_1\rangle)\,, \tag{68}$$

$$T_4{}^{(1)}=-\frac{1}{n}\int\left(\frac{\partial F_{1p}}{\partial\zeta_1}+\frac{3}{\zeta_1}F_{1p}\right)d\zeta_4\,, \tag{69}$$

$$Z_5{}^{*(1)}=-\left\langle\frac{\partial F_1}{\partial\zeta_2}\right\rangle=-\frac{\partial F_1{}^*}{\partial\zeta_2}\,, \tag{70}$$

$$T_5{}^{(1)}=-\frac{1}{n}\int\frac{\partial F_{1p}}{\partial\zeta_2}d\zeta_4\,. \tag{71}$$

Introducing $S_1(\zeta_1,\zeta_2,z_3,\zeta_4,\zeta_5,-)$ by

$$S_1=\frac{1}{n}\int F_{1p}d\zeta_4\,, \tag{72}$$

we have

$$T_j{}^{(1)}=\frac{\partial S_1}{\partial\zeta_{3+j}}\,, \quad T_{3+j}^{(1)}=-\frac{\partial S_1}{\partial\zeta_j}\,, \qquad (j=1,2)\,. \tag{73}$$

Therefore the results given by equations (64) to (71) are the same as those given

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in the canonical case (HORI 1966).

It is to be noted that $T_1^{(1)}$ given by equation (65) is not a unique choice for $T_1^{(1)}$. By another choice of the integration constant we may have

$$T_1^{(1)}=\frac{1}{n}F_1 . \tag{74}$$

Equation (68) is then changed to

$$\begin{aligned} Z_4^{*(1)}&=-\left\langle\frac{\partial F_1}{\partial\zeta_1}+\frac{3\mu^2}{\zeta_1^4}T_1^{(1)}\right\rangle=-\left\langle\frac{\partial F_1}{\partial\zeta_1}+\frac{3}{\zeta_1}F_1\right\rangle \\ &=-\left(\frac{\partial F_1^*}{\partial\zeta_1}+\frac{3}{\zeta_1}F_1^*\right)=0 . \end{aligned} \tag{75}$$

Obviously, the $T_1^{(1)}$ given by equation (74) cannot be obtained from the determining function S_1 of equation (72), so the choice of equation (74) leads to another form of the solution where the perturbations in z_j are not obtained from a single determining function.

In the second approximation, we have, according to equations (29),

$$\left.\begin{aligned} [Z^{(1)}+Z^{*(1)},\ T^{(1)}]_j&=\frac{\partial}{\partial\zeta_{3+j}}\{F_1+F_1^*,\ S_1\} , \\ [Z^{(1)}+Z^{*(1)},\ T^{(1)}]_{3+j}&=-\frac{\partial}{\partial\zeta_j}\{F_1+F_1^*,\ S_1\} , \end{aligned}\right\}\ (j=1,2) ,$$

and $Z_j^{*(2)}$ and $T_j^{(2)}$ are obtained exactly the same as in the canonical case.

The new equations of motion are

$$\frac{d\zeta_j}{dt}=Z_j^{*(1)}(\zeta_1,\zeta_2,z_3,-,-,-)+Z_j^{*(2)}(\zeta_1,\zeta_2,z_3,-,\zeta_5,-) ,\quad (j=2,5) , \tag{76}$$

$$\frac{d\zeta_4}{dt}=\frac{\mu^2}{\zeta_1^3}+Z_4^{*(1)}+Z_4^{*(2)} , \tag{77}$$

with

$$\zeta_1=\text{const.} \tag{78}$$

However, as mentioned above, equation (77) is separated from equations (76) and need not be considered until the latter are solved.

In equations (76), $Z_j^{*(1)}$ are the lowest order terms and play the role of the undisturbed forces. Since $Z_j^{*(1)}$ are free from all the angular variables, the solution of equations (76) is attained in the same way as the preceding step. We consider a second transformation $\zeta_2,\zeta_5\to\zeta_2',\zeta_5'$ in the form

$$\zeta_j=\zeta_j'+T_j^{*(1)}(\zeta_1,\zeta_2',z_3,-,\zeta_5',-)+\cdots ,\quad (j=2,5) , \tag{79}$$

ζ_1 as well as z_3 being considered as a parameter. The auxiliary equations are

$$\frac{d\zeta_j'}{d\tau^*}=Z_j^{**(1)} ,\qquad (j=2,5) \tag{80}$$

with

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$$Z_j^{**(1)}=Z_j^{*(1)}(\zeta_1, \zeta_2', z_3, -, -, -)\,, \qquad (j=2, 5)\,. \tag{81}$$

In view of equations (66) and (70), we have

$$\zeta_2'=\text{const. in } \tau^*\,, \tag{82}$$

and

$$\zeta_5'=Z_5^{**(1)}\cdot\tau^*+\text{const.} \tag{83}$$

The first order equations (35′) are now written as

$$-\frac{dT_j^{*(1)}}{d\tau^*}+Z_j^{*(2)}(\zeta_1, \zeta_2', z_3, -, \zeta_5', -)=Z_j^{**(2)}\,, \qquad (j=2, 5)\,. \tag{84}$$

The guiding principle which determines $Z_j^{**(2)}$ and $T_j^{*(1)}$ is that the $T_j^{*(1)}$ are free from secular or mixed secular terms in τ^* or, which is the same, in ζ_5'. Thus equations (84) lead to

$$Z_j^{**(2)}=\langle Z_j^{*(2)}\rangle\,, \tag{85}$$

and

$$T_j^{*(1)}=(Z_5^{**(1)})^{-1}\int(Z_j^{*(2)}-Z_j^{**(2)})d\zeta_5'\,, \tag{86}$$

where $\langle\ \rangle$ denotes the average with respect to ζ_5'.

The new equations of motion become

$$\frac{d\zeta_2'}{dt}=0\,, \tag{87}$$

$$\frac{d\zeta_5'}{dt}=Z_5^{**(1)}(\zeta_1, \zeta_2', z_3, -, -, -)+Z_5^{**(2)}(\zeta_1, \zeta_2', z_3, -, -, -)\,, \tag{88}$$

and their solution is

$$\zeta_2'=\text{const.}\,, \tag{89}$$

$$\zeta_5'=(Z_5^{**(1)}+Z_5^{**(2)})t+\text{const.} \tag{90}$$

Since equations (76) are solved, we next consider equation (77). Applying transformation (79) to the right-hand member of equation (77), we have

$$\begin{aligned}\frac{d\zeta_4}{dt}=&\frac{\mu^2}{\zeta_1^3}+Z_4^{*(1)}(\zeta_1, \zeta_2', z_3, -, -, -)\\&+\frac{\partial Z_4^{*(1)}}{\partial\zeta_2'}T_2^{*(1)}+Z_4^{*(2)}(\zeta_1, \zeta_2', z_3, -, \zeta_5', -)\end{aligned} \tag{91}$$

up to the second order, or after integration,

$$\begin{aligned}\zeta_4=&\left(\frac{\mu^2}{\zeta_1^3}+Z_4^{*(1)}+\langle Z_4^{*(2)}\rangle\right)t+\text{const.}\\&+(Z_5^{**(1)})^{-1}\int\left(\frac{\partial Z_4^{*(1)}}{\partial\zeta_2'}T_2^{*(1)}+Z_4^{*(2)}-\langle Z_4^{*(2)}\rangle\right)d\zeta_5'\,.\end{aligned} \tag{92}$$

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The first part of the right-hand member is the secular perturbation up to the second order and is denoted by ζ_4', while the second part is the long period perturbation of the first order and is denoted by $T_4^{*(1)}$.

Finally, the perturbations in z_6 are obtained from

$$\frac{dz_6}{dt}=Z_6^{(1)}(z_1, z_2, z_3, z_4, z_5, -)+Z_6^{(2)}(z_1, z_2, z_3, z_4, z_5, -) . \tag{93}$$

In fact the transformations are written, to the first order, in the form

$$z_1=\zeta_1+T_1^{(1)}(\zeta_1, \zeta_2', z_3, \zeta_4', \zeta_5', -) ,$$

$$z_j=\zeta_j'+T_j^{(1)}(\zeta_1, \zeta_2', z_3, \zeta_4', \zeta_5' -)+T_j^{*(1)}(\zeta_1, \zeta_2', z_3, -, \zeta_5', -) , \quad (j=2, 4, 5) ,$$

and applying them to the right-hand member of equation (93), we have

$$\frac{dz_6}{dt}=Z_6^{(1)}(\zeta_1, \zeta_2', z_3, \zeta_4', \zeta_5', -)+\frac{\partial Z_6^{(1)}}{\partial \zeta_1} T_1^{(1)}$$
$$+\sum_{j=2,4,5} \frac{\partial Z_6^{(1)}}{\partial \zeta_j'} (T_j^{(1)}+T_j^{*(1)})+Z_6^{(2)}(\zeta_1, \zeta_2', z_3, \zeta_4', \zeta_5', -) , \tag{94}$$

an equation which can be integrated to give z_6, since the right-hand member of equation (94) is now a known function of t. Omitting the ζ_4'-dependent terms of second order, z_6 consists of the short and long-period terms of the first order and secular terms of the second order, being the same as obtained in the canonical case.

The above-stated solution of the satellite motion is along the general procedure of Section 5. Alternatively we can proceed exactly as in the canonical case, where the problem is of the sixth order, and the transformations of z_6 and ζ_4 are taken into account in the first stage $z\to\zeta$ and in the second stage $\zeta\to\zeta'$, respectively. However, the amount of calculation seems to be the same by both procedures.

7. *Application to Non-linear Oscillations*

We consider as an illustration the non-linear oscillations given by van del Pol's equation

$$\ddot{x}+\varepsilon(x^2-1)\dot{x}+x=0 , \tag{95}$$

or

$$\left.\begin{aligned} \dot{z}_1&=z_2 , \\ \dot{z}_2&=-z_1-\varepsilon(z_1^2-1)z_2 . \end{aligned}\right\} \tag{96}$$

We have

$$Z_1^{(0)}(z)=z_2 , \quad Z_2^{(0)}(z)=-z_1 , \tag{97}$$

$$Z_1^{(1)}(z)=0 , \quad Z_2^{(1)}(z)=-\varepsilon(z_1^2-1)z_2 . \tag{98}$$

The auxiliary equations (39) read

$$\frac{d\zeta_1}{d\tau}=Z_1^{(0)}=\zeta_2 , \quad \frac{d\zeta_2}{d\tau}=Z_2^{(0)}=-\zeta_1 , \tag{99}$$

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with the solution

$$\zeta_1 = c\cos\tau\,, \qquad \zeta_2 = -c\sin\tau\,. \tag{100}$$

In the first order theory, $T_j^{(1)}$, $Z_j^{*(1)}$ are determined from equation (35′), or in view of equation (41) and equations (97), (98),

$$\left.\begin{aligned} -\frac{dT_1^{(1)}}{d\tau} + T_2^{(1)} &= Z_1^{*(1)}\,, \\ -\frac{dT_2^{(1)}}{d\tau} - T_1^{(1)} + Z_2^{(1)} &= Z_2^{*(1)}\,, \end{aligned}\right\} \tag{101}$$

where $Z_2^{(1)}$ are known functions of τ by equations (100). Therefore equations (101) are linear equations with constant coefficients. This is the general feature when the undisturbed oscillation is given by $\ddot{x}+x=0$, and the same feature appears in higher orders.

From equations (101) we have

$$\frac{d^2T_1^{(1)}}{d\tau^2} + T_1^{(1)} = Z_2^{(1)} - Z_2^{*(1)} - \frac{dZ_1^{*(1)}}{d\tau}\,. \tag{102}$$

$Z_2^{(1)}$, which is $\varepsilon(\zeta_2-\zeta_1^2\zeta_2)$ in the present case, is expressed in general by Fourier series with multiples of τ as arguments. If $\cos\tau$ or $\sin\tau$ is present in the right-hand side of equation (102), $T_1^{(1)}$ has a mixed secular term, which should be avoided, according to the principle stated in Section 4. Such terms of $Z_2^{(1)}$ being denoted by $[Z_2^{(1)}]_s$, equation (102) gives

$$\frac{d^2T_1^{(1)}}{d\tau^2} + T_1^{(1)} = Z_2^{(1)} - [Z_2^{(1)}]_s \tag{103}$$

and

$$\frac{dZ_1^{*(1)}}{d\tau} + Z_2^{*(1)} = [Z_2^{(1)}]_s\,, \tag{104}$$

the latter being satisfied by the choice

$$\frac{dZ_1^{*(1)}}{d\tau} = Z_2^{*(1)} = \frac{1}{2}[Z_2^{(1)}]_s\,. \tag{105}$$

Since $[Z_2^{(1)}]_s$ consists of sine or cosine functions of the single multiple of τ, condition (105) implies that $Z_1^{*(1)}$ changes into $Z_2^{*(1)}$ if ζ_1 and ζ_2 in $Z_1^{*(1)}$ are changed to ζ_2 and $-\zeta_1$, respectively. Generally, we determine $Z_1^{*(n)}$ and $Z_2^{*(n)}$ by

$$\frac{dZ_1^{*(n)}}{d\tau} = Z_2^{*(n)}\,, \qquad (n=2, 3, \cdots)\,, \tag{106}$$

as equations (49) with $j=1$ suggest. Owing to conditions (105) and (106), the transformed equations of motion become integrable as shown below.

In the present case, $Z_2^{(1)}$ consists of $\varepsilon\zeta_2$ and $-\varepsilon\zeta_1^2\zeta_2$, and each is considered separately since equations (103) and (105) are linear. Consider, in general, the term $\zeta_1^i\zeta_2^j$. Then $[\zeta_1^i\zeta_2^j]_s$ determines $Z_2^{*(1)}$, $Z_1^{*(1)}$ by equation (105), and $T_1^{(1)}$ is given by integration of equation (103). We write the result in the form

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$$T_1^{(1)}=[\zeta_1^i\zeta_1^j]_p\ . \tag{107}$$

In the Appendix, $[\zeta_1^i\zeta_2^j]_s$ and $[\zeta_1^i\zeta_2^j]_p$ are tabulated for $i+j=2, 3, \cdots, 8$, $i\geqq j$. The cases $i<j$ are obtained by the change $\zeta_1\to\zeta_2$, $\zeta_2\to-\zeta_1$ corresponding to $\tau\to\tau+\pi/2$. With the help of the table we can manipulate the calculation without recourse to explicit use of τ.

With $Z_2^{(1)}=\varepsilon(\zeta_2-\zeta_1^2\zeta_2)$ we have

$$Z_2^{*(1)}=\frac{\varepsilon}{2}\left(1-\frac{c^2}{4}\right)\zeta_2,\quad Z_1^{*(1)}=\frac{\varepsilon}{2}\left(1-\frac{c^2}{4}\right)\zeta_1,\quad c^2=\zeta_1^2+\zeta_2^2\ , \tag{108}$$

$$T_1^{(1)}=\frac{\varepsilon}{32}(3\zeta_1^2\zeta_2-\zeta_2^3),\quad T_2^{(1)}=\frac{\varepsilon}{32}(16\zeta_1-7\zeta_1^3+5\zeta_1\zeta_2^2)\ , \tag{109}$$

where $T_2^{(1)}$ is obtained from the first of equations (101) in view of

$$\frac{d}{d\tau}=\zeta_2\frac{\partial}{\partial\zeta_1}-\zeta_1\frac{\partial}{\partial\zeta_2}\ .$$

In the second approximation, $Z_j^{*(2)}$ and $T_j^{(2)}$ are determined by equation (36′), or equivalently by

$$\left.\begin{aligned}-\frac{dT_1^{(2)}}{d\tau}+T_2^{(2)}+\Phi_1^{(2)}&=Z_1^{*(2)}\ ,\\ -\frac{dT_2^{(2)}}{d\tau}-T_1^{(2)}+\Phi_2^{(2)}&=Z_2^{*(2)}\ ,\end{aligned}\right\} \tag{110}$$

where

$$\left.\begin{aligned}\Phi_1^{(2)}&=\frac{1}{2}\left[T_1^{(1)}\frac{\partial Z_1^{*(1)}}{\partial\zeta_1}+T_2^{(1)}\frac{\partial Z_1^{*(1)}}{\partial\zeta_2}-Z_1^{*(1)}\frac{\partial T_1^{(1)}}{\partial\zeta_1}-(Z_2^{(1)}+Z_2^{*(1)})\frac{\partial T_1^{(1)}}{\partial\zeta_2}\right],\\ \Phi_2^{(2)}&=\frac{1}{2}\left[T_1^{(1)}\frac{\partial}{\partial\zeta_1}(Z_2^{(1)}+Z_2^{*(1)})+T_2^{(1)}\frac{\partial}{\partial\zeta_2}(Z_2^{(1)}+Z_2^{*(1)})-Z_1^{*(1)}\frac{\partial T_2^{(1)}}{\partial\zeta_1}\right.\\ &\qquad\left.-(Z_2^{(1)}+Z_2^{*(1)})\frac{\partial T_2^{(1)}}{\partial\zeta_2}\right].\end{aligned}\right\} \tag{111}$$

Analogous to equations (103) and (105), we have

$$\frac{d^2T_1^{(2)}}{d\tau^2}+T_1^{(2)}=\frac{d\Phi_1^{(2)}}{d\tau}+\Phi_2^{(2)}-\left[\frac{d\Phi_1^{(2)}}{d\tau}+\Phi_2^{(2)}\right]_s, \tag{112}$$

$$\frac{dZ_1^{*(2)}}{d\tau}=Z_2^{*(2)}=\frac{1}{2}\left[\frac{d\Phi_1^{(2)}}{d\tau}+\Phi_2^{(2)}\right]_s. \tag{113}$$

In view of equation (107) we can write

$$T_1^{(2)}=\left[\frac{d\Phi_1^{(2)}}{d\tau}+\Phi_2^{(2)}\right]_p. \tag{114}$$

Then $T_2^{(2)}$ is determined from the first of equations (110). The result is

$$\frac{d\Phi_1^{(2)}}{d\tau}+\Phi_2^{(2)}=\frac{\varepsilon^2}{128}(32\zeta_1-12\zeta_1^3-92\zeta_1\zeta_2^2+\zeta_1^5+58\zeta_1^3\zeta_2^2-7\zeta_1\zeta_2^4)\ ,$$

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and by the table in the Appendix,

$$Z_1{}^{*(2)}=-\frac{\varepsilon^2}{256}(32-23c^2+7c^4)\zeta_2,\quad Z_2{}^{*(2)}=\frac{\varepsilon^2}{256}(32-32c^2+7c^4)\zeta_1\,, \tag{115}$$

$$\left.\begin{aligned} T_1{}^{(2)}&=\frac{\varepsilon^2}{1536}(-30\zeta_1{}^3+90\zeta_1\zeta_2{}^2+5\zeta_1{}^5-26\zeta_1{}^3\zeta_2{}^2+\zeta_1\zeta_2{}^4)\,,\\ T_2{}^{(2)}&=\frac{\varepsilon^2}{1536}(-192\zeta_2+162\zeta_1{}^2\zeta^2+186\zeta_2{}^3-79\zeta_1{}^4\zeta_2-82\zeta_1{}^2\zeta_2{}^3-35\zeta_2{}^5)\,. \end{aligned}\right\} \tag{116}$$

Similarly we have, in the third approximation,

$$\begin{aligned} \frac{d\Phi_1{}^{(3)}}{d\tau}+\Phi_2{}^{(3)}=\frac{\varepsilon^3}{3072}(&-420\zeta_1{}^2\zeta_2-468\zeta_2{}^3+506\zeta_1{}^4\zeta_2+1096\zeta_1{}^2\zeta_2{}^3+102\zeta_2{}^5-151\zeta_1{}^6\zeta_2\\ &-201\zeta_1{}^4\zeta_2{}^3-121\zeta_1{}^2\zeta_2{}^5-7\zeta_2{}^7)\,, \end{aligned}$$

$$\left.\begin{aligned} Z_1{}^{*(3)}&=\frac{\varepsilon^3}{12288}(-912c^2+528c^4-69c^6)\zeta_1\,,\\ Z_2{}^{*(3)}&=\frac{\varepsilon^3}{12288}(-912c^2+528c^4-69c^6)\zeta_2\,, \end{aligned}\right\} \tag{117}$$

limiting to only the secular perturbations. If periodic perturbations of the third order in x are required, we need $T_1{}^{(3)}$ which is

$$\begin{aligned} T_1{}^{(3)}=\frac{\varepsilon^3}{147456}(&-216\zeta_1{}^2\zeta_2+72\zeta_2{}^3-2062\zeta_1{}^4\zeta_2-2188\zeta_1{}^2\zeta_2{}^3+850\zeta_2{}^5\\ &+589\zeta_1{}^6\zeta_2+835\zeta_1{}^4\zeta_2{}^3+135\zeta_1{}^2\zeta_2{}^5-175\zeta_2{}^7)\,. \end{aligned} \tag{118}$$

The new equations of motion are, by equations (108), (115), and (117),

$$\left.\begin{aligned} \frac{d\zeta_1}{dt}&=\sum Z_1{}^{*(j)}=\zeta_2+\frac{\varepsilon}{2}\left(1-\frac{c^2}{4}\right)\zeta_1-\frac{\varepsilon^2}{256}(32-32c^2+7c^4)\zeta_2\\ &\qquad+\frac{\varepsilon^3}{12288}(-912c^2+528c^4-69c^6)\zeta_1\,,\\ \frac{d\zeta_2}{dt}&=\sum Z_2{}^{*(j)}=-\zeta_1+\frac{\varepsilon}{2}\left(1-\frac{c^2}{4}\right)\zeta_2+\frac{\varepsilon^2}{256}(32-32c^2+7c^4)\zeta_1\\ &\qquad+\frac{\varepsilon^3}{12288}(-912c^2+528c^4-69c^6)\zeta_2\,. \end{aligned}\right\} \tag{119}$$

Hence

$$2\left(\zeta_1\frac{d\zeta_1}{dt}+\zeta_2\frac{d\zeta_2}{dt}\right)=\frac{dc^2}{dt}=\varepsilon\left(1-\frac{c^2}{4}\right)c^2+\frac{\varepsilon^3}{6144}(-912c^2+528c^4-69c^6)c^2\,, \tag{120}$$

and

$$\frac{1}{c^2}\left(\zeta_2\frac{d\zeta_1}{dt}-\zeta_1\frac{d\zeta_2}{dt}\right)=\frac{d}{dt}\tan^{-1}\frac{\zeta_1}{\zeta_2}=1-\frac{\varepsilon^2}{256}(33-32c^2+7c^4)\,. \tag{121}$$

Then $c^2=\zeta_1{}^2+\zeta_2{}^2$ and ζ_1/ζ_2 are obtained by two quadratures with two constants of integration. In fact, equation (120) leads to

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$$c^2 e^{(23/128)\varepsilon^2 c^2}\left|1+\frac{\varepsilon^2}{16}-\frac{c^2}{4}\right|^{-1+\varepsilon^2/16}=\text{const. } e^{\varepsilon t}\ , \tag{122}$$

in the approximation neglecting $O(\varepsilon^4)$. This equation shows $c^2 \to 4+\varepsilon^2/4$ $(\varepsilon t \to +\infty)$ (the limit cycle), of which we note that the ε^2 contribution comes from the third order theory.

After ζ_1 and ζ_2 are obtained, $z_1(=x)$ and $z_2(=\dot{x})$ are given by equation (13) or

$$z_j=\zeta_j+T_j^{(1)}+T_j^{(2)}+\frac{1}{2}\left(T_1^{(1)}\frac{\partial T_j^{(1)}}{\partial \zeta_1}+T_2^{(1)}\frac{\partial T_j^{(1)}}{\partial \zeta_2}\right), \quad (j=1, 2)\ , \tag{123}$$

or, in view of equations (109) and (116),

$$\left.\begin{aligned} z_1&=\zeta_1+\frac{\varepsilon}{32}(3\zeta_1^2\zeta_2-\zeta_2^3)+\frac{\varepsilon^2}{6144}(24\zeta_1^3+216\zeta_1\zeta_2^2-43\zeta_1^5+58\zeta_1^3\zeta_2^2-59\zeta_1\zeta_2^4)\ ,\\ z_2&=\zeta_2+\frac{\varepsilon}{32}(16\zeta_1-7\zeta_1^3+5\zeta_1\zeta_2^2)\\ &\quad+\frac{\varepsilon^2}{6144}(-768\zeta_2+1272\zeta_1^2\zeta_2+696\zeta_2^3-715\zeta_1^4\zeta_2-70\zeta_1^2\zeta_2^3-155\zeta_2^5)\ . \end{aligned}\right\} \tag{124}$$

Equations (119) admit the special solution

$$\zeta_1=c\cos(\theta+\varphi)\ , \quad \varphi_2=-c\sin(\theta+\varphi)\ , \tag{125}$$

with

$$c^2=4+\frac{\varepsilon^2}{4}\ , \quad \theta=\left(1-\frac{\varepsilon^2}{16}\right)t\ , \tag{126}$$

and a phase constant φ. Let φ be determined so that z_2 vanishes at $t=0$. Equations (123) give the inverse transformation

$$\zeta_j=z_j-\bar{T}_j^{(1)}-\bar{T}_j^{(2)}+\frac{1}{2}\left(\bar{T}_1^{(1)}\frac{\partial \bar{T}_j^{(1)}}{\partial z_1}+\bar{T}_2^{(1)}\frac{\partial \bar{T}_j^{(1)}}{\partial z_2}\right), \quad (j=1, 2)\ , \tag{127}$$

or

$$\left.\begin{aligned} \zeta_1&=z_1-\frac{\varepsilon}{32}(3z_1^2z_2-z_2^3)+\frac{\varepsilon^2}{6144}(264z_1^3-504z_1z_2^2-83z_1^5+266z_1^3z_2-67z_1z_2^4)\ ,\\ \zeta_2&=z_2-\frac{\varepsilon}{32}(16z_1-7z_1^3+5z_1z_2^2)\\ &\quad+\frac{\varepsilon^2}{6144}(768z_2-24z_1^2z_2+792z_2^3-83z_1^4z_2+586z_1^2z_2^3+125z_2^5)\ . \end{aligned}\right\} \tag{128}$$

With $t=0$ we have

$$\left.\begin{aligned} \zeta_1(0)&=z_1(0)+\frac{\varepsilon^2}{6144}(264z_1^3(0)-83z_1^5(0))\ ,\\ \zeta_2(0)&=-\frac{\varepsilon}{32}(16z_1(0)-7z_1^3(0))\ , \end{aligned}\right\} \tag{129}$$

hence the condition

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$$\zeta_1^2(0)+\zeta_2^2(0)=c^2=4+\frac{\varepsilon^2}{4}$$

gives $z_1(0)=2+\varepsilon^2/96$, and therefore, equations (129) become

$$\zeta_1(0)=2\left(1+\frac{\varepsilon^2}{32}\right)\cos\varphi=2-\frac{15}{192}\varepsilon^2\,, \quad \text{or,} \quad \cos\varphi=1-\frac{9}{128}\varepsilon^2\,,$$

$$\zeta_2(0)=-2\left(1+\frac{\varepsilon^2}{32}\right)\sin\varphi=\frac{3}{4}\varepsilon\,, \quad \text{or,} \quad \sin\varphi=-\frac{3}{8}\varepsilon\,,$$

so that

$$\varphi=-\frac{3}{8}\varepsilon+O(\varepsilon^3)\,. \tag{130}$$

Finally, we find the well-known expressions for the limit cycle,

$$\left.\begin{aligned} x=z_1&=\left(2-\frac{\varepsilon^2}{8}\right)\cos\theta+\frac{3}{4}\varepsilon\sin\theta-\frac{\varepsilon}{4}\sin 3\theta+\frac{3}{16}\varepsilon^2\cos 3\theta-\frac{5}{96}\varepsilon^2\cos 5\theta\,,\\ \dot{x}=z_2&=\left(-2+\frac{\varepsilon^2}{4}\right)\sin\theta+\frac{3}{4}\varepsilon\cos\theta-\frac{3}{4}\varepsilon\cos 3\theta-\frac{9}{16}\varepsilon^2\sin 3\theta+\frac{25}{96}\varepsilon^2\sin 5\theta\,.\end{aligned}\right\} \tag{131}$$

We note that

$$x=(2+a\varepsilon^2)\cos\theta+\frac{3}{4}\varepsilon\sin\theta-\frac{\varepsilon}{4}\sin 3\theta+\frac{3}{16}\varepsilon^2\cos 3\theta-\frac{5}{96}\varepsilon^2\cos 5\theta\,, \tag{132}$$

with an arbitrary constant a, satisfies equation (95) if terms of $O(\varepsilon^3)$ are neglected.

8. *Generalized Canonical System*

In concluding the present article, we remark that when the original theory (HORI 1966) for canonical cases is applied to the generalized canonical system (6), with Hamiltonian (7), and if its algorithm is required to be independent of any knowledge about the momenta p artificially introduced, this algorithm becomes exactly the same as that presented so far.

In fact, let $z, p\to\zeta, \pi$ be the canonical transformation applied to the generalized canonical system. Furthermore, let $S(\zeta, \pi)$ be the determining function of the canonical transformation, and let $F^*(\zeta, \pi)$ be the new Hamiltonian, so that the new equations of motion are

$$\frac{d\zeta_j}{dt}=\frac{\partial F^*}{\partial\pi_j}, \quad \frac{d\pi_j}{dt}=-\frac{\partial F^*}{\partial\zeta_j}\,, \qquad (j=1,\cdots,m)\,. \tag{133}$$

We choose S and F^* in the form

$$S=\pi_j T_j(\zeta)\,, \quad F^*=\pi_j Z_j^*(\zeta)\,. \tag{134}$$

First we prove

$$D_S^n F(\zeta, \pi)=\pi_j \mathscr{D}_T^n Z_j(\zeta)\,. \tag{135}$$

In fact, the form of F

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$$F(\zeta, \pi)=\pi_j Z_j(\zeta) , \tag{136}$$

as well as that of S leads to

$$D_S F=\{F, S\}=\{\pi_j Z_j, \pi_j T_j\}=\pi_j[Z, T]_j=\pi_j \mathscr{D}_T Z_j . \tag{137}$$

If we assume equation (135) for $n=N-1$, we have

$$\begin{aligned} D_S^N F=D_S D_S^{N-1} F&=\{D_S^{N-1} F, S\} \\ &=\{\pi_j \mathscr{D}_T^{N-1} Z_j, \pi_j T_j\} \\ &=\pi_j\left(\frac{\partial}{\partial \zeta_k} \mathscr{D}_T^{N-1} Z_j\right) T_k-(\mathscr{D}_T^{N-1} Z_k)\pi_j \frac{\partial T_j}{\partial \zeta_k} \\ &=\pi_j\left[T_k \frac{\partial}{\partial \zeta_k} \mathscr{D}_T^{N-1} Z_j-(\mathscr{D}_T^{N-1} Z_k) \frac{\partial T_j}{\partial \zeta_k}\right] \\ &=\pi_j \mathscr{D}_T(\mathscr{D}_T^{N-1} Z_j)=\pi_j \mathscr{D}_T^N Z_j , \end{aligned} \tag{138}$$

thus equation (135) is proved by mathematical induction. Hence we have

$$\pi_j Z_j^*=F^*=e^{D_S} F(\zeta, \pi)=\sum \frac{1}{n!} D_S^n F=\pi_j \sum \frac{1}{n!} \mathscr{D}_T^n Z_j=\pi_j e^{\mathscr{D}_T} Z_j$$

or, if these hold with arbitrary π, we have

$$Z_j^*=e^{\mathscr{D}_T} Z_j . \tag{139}$$

This furnishes another proof of equation (28).

Further, the requirement

$$\{F_0, F^*\}=\{\pi_j Z_j^{(0)}, \pi_j Z_j^*\}=0 \tag{140}$$

yields

$$0=\pi_j \frac{\partial Z_j^{(0)}}{\partial \zeta_k} Z_k^*-Z_k^{(0)} \pi_j \frac{\partial Z_j^*}{\partial \zeta_k}=\pi_j\left[Z_k^* \frac{\partial Z_j^{(0)}}{\partial \zeta_k}-Z_k^{(0)} \frac{\partial Z_j^*}{\partial \zeta_k}\right]=\pi_j[Z^{(0)}, Z^*]_j , \tag{141}$$

or the tractability condition (47), when $\{F_0, F^*\}=0$ is required to hold with arbitrary π.

The remarks in Section 8 are partly due to the comments of H. Kinoshita, and the author thanks him for his comments as well as for other discussions on this subject.

Appendix. *Table of* $[\zeta_1^i \zeta_2^j]_s$ *and* $[\zeta_1^i \zeta_2^j]_p$.

$\zeta_1^i \zeta_2^j$	$[\zeta_1^i \zeta_2^j]_s$	$[\zeta_1^i \zeta_2^j]_p$
ζ_1	ζ_1	
ζ_1^2		$\frac{1}{3}(\zeta_1^2+2\zeta_2^2)$
$\zeta_1\zeta_2$		$-\frac{1}{3}\zeta_1\zeta_2$

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$\zeta_1^i\zeta_2^j$	$[\zeta_1^i\zeta_2^j]_s$	$[\zeta_1^i\zeta_2^j]_p$
ζ_1^3	$\frac{3}{4}c^2\zeta_1$	$\frac{1}{32}(-\zeta_1^3+3\zeta_1\zeta_2^2)$
$\zeta_1^2\zeta_2$	$\frac{1}{4}c^2\zeta_2$	$\frac{1}{32}(-3\zeta_1^2\zeta_2+\zeta_2^3)$
ζ_1^4		$\frac{1}{15}(3\zeta_1^4+12\zeta_1^2\zeta_2^2+8\zeta_2^4)$
$\zeta_1^3\zeta_2$		$-\frac{1}{15}(3\zeta_1^3\zeta^2+2\zeta_1\zeta_2^3)$
$\zeta_1^2\zeta_2^2$		$\frac{1}{15}(2\zeta_1^4+3\zeta_1^2\zeta_2^2+2\zeta_2^4)$
ζ_1^5	$\frac{5}{8}c^4\zeta_1$	$\frac{1}{48}(-2\zeta_1^5+5\zeta_1^3\zeta_2^2+5\zeta_1\zeta_2^4)$
$\zeta_1^4\zeta_2$	$\frac{1}{8}c^4\zeta_2$	$\frac{1}{48}(-4\zeta_1^4\zeta_2-\zeta_1^2\zeta_2^3+\zeta_2^5)$
$\zeta_1^3\zeta_2^2$	$\frac{1}{8}c^4\zeta_1$	$\frac{1}{96}(\zeta_1^5-4\zeta_1^3\zeta_2^2-\zeta_1\zeta_2^4)$
ζ_1^6		$\frac{1}{35}(5\zeta_1^6+30\zeta_1^4\zeta_2^2+40\zeta_1^2\zeta_2^4+16\zeta_2^6)$
$\zeta_1^5\zeta_2$		$-\frac{1}{105}(15\zeta_1^5\zeta_2+20\zeta_1^3\zeta_2^3+8\zeta_1\zeta_2^5)$
$\zeta_1^4\zeta_2^2$		$\frac{1}{105}(6\zeta_1^6+15\zeta_1^4\zeta_2^2+20\zeta_1^2\zeta_2^4+8\zeta_2^6)$
$\zeta_1^3\zeta_2^3$		$-\frac{1}{105}(6\zeta_1^5\zeta_2+15\zeta_1^3\zeta_2^3+6\zeta_1\zeta_2^5)$
ζ_1^7	$\frac{35}{64}c^6\zeta_1$	$\frac{1}{3072}(-141\zeta_1^7+273\zeta_1^5\zeta_2^2+665\zeta_1^3\zeta_2^4+315\zeta_1\zeta_2^6)$
$\zeta_1^6\zeta_2$	$\frac{5}{64}c^6\zeta_2$	$\frac{1}{3072}(-219\zeta_1^6\zeta_2-185\zeta_1^4\zeta_2^3+15\zeta_1^2\zeta_2^5+45\zeta_2^7)$
$\zeta_1^5\zeta_2^2$	$\frac{5}{64}c^6\zeta_1$	$\frac{1}{3072}(13\zeta_1^7-81\zeta_1^5\zeta_2^2-25\zeta_1^3\zeta_2^4+5\zeta_1\zeta_2^6)$
$\zeta_1^4\zeta_2^3$	$\frac{3}{64}c^6\zeta_2$	$\frac{1}{3072}(-37\zeta_1^6\zeta_2-135\zeta_1^4\zeta_2^3-15\zeta_1^2\zeta_2^5+19\zeta_2^7)$
ζ_1^8		$\frac{1}{315}(35\zeta_1^8+280\zeta_1^6\zeta_2^2+560\zeta_1^4\zeta_2^4+448\zeta_1^2\zeta_2^6+128\zeta_2^8)$
$\zeta_1^7\zeta_2$		$-\frac{1}{315}(35\zeta_1^7\zeta_2+70\zeta_1^5\zeta_2^3+40\zeta_1^3\zeta_2^5+16\zeta_1\zeta_2^7)$
$\zeta_1^6\zeta_2^2$		$\frac{1}{315}(10\zeta_1^8+35\zeta_1^6\zeta_2^2+70\zeta_1^4\zeta_2^4+56\zeta_1^2\zeta_2^6+16\zeta_2^8)$
$\zeta_1^5\zeta_2^3$		$-\frac{1}{315}(10\zeta_1^7\zeta_2+35\zeta_1^5\zeta_2^3+44\zeta_1^3\zeta_2^5+8\zeta_1\zeta_2^7)$
$\zeta_1^4\zeta_2^4$		$\frac{1}{315}(8\zeta_1^8+28\zeta_1^6\zeta_2^2+35\zeta_1^4\zeta_2^4+28\zeta_1^2\zeta_2^6+8\zeta_2^8)$

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